

Technology Stack

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript, Bootstrap	Provides a responsive and user-friendly interface for applicants to enter loan details and view eligibility results.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3.10+, Flask Framework	Lightweight Flask API controller handles web requests, user authentication sessions, and calls the ML model to predict loan eligibility.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	MySQL, SQLAlchemy ORM	Relational storage maps user details, loan applications, prediction history, and results securely.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Local server, Render / Heroku (Optional)	Used for development, testing, and deployment of the Smart Lender application.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub	Collaborative version control and branch merging.
6	Machine Learning & Other Tools (e.g., Scikit-learn, Pandas, TensorFlow)	Scikit-learn, Pandas, NumPy, Pickle, Matplotlib	Scikit-learn builds the loan eligibility prediction model, Pandas & NumPy handle data preprocessing, Pickle stores the trained model, and Matplotlib generates performance visualizations.